Crossing symmetry

We have for Bhabha scattering

$$\langle|\mathcal{M}|^2\rangle = 2\alpha^2\left(\frac{s^2+u^2}{t^2}+\frac{2u^2}{st}+\frac{t^2+u^2}{s^2}\right)$$

and for Moller scattering

$$\langle|\mathcal{M}|^2\rangle = 2\alpha^2\left(\frac{s^2+u^2}{t^2}+\frac{2s^2}{tu}+\frac{s^2+t^2}{u^2}\right)$$

Bhabha and Moller scattering share $s \leftrightarrow u$ crossing symmetry.

We have for electron-positron annihilation

$$\langle|\mathcal{M}|^2\rangle = 2\alpha^2\left(\frac{u}{t}+\frac{t}{u}\right)$$

and for Compton scattering

$$\langle|\mathcal{M}|^2\rangle = -2\alpha^2\left(\frac{u}{s}+\frac{s}{u}\right)$$

Annihilation and Compton scattering share $s \leftrightarrow t$ crossing symmetry. The minus sign is the result of moving a fermion from inbound (in annihilation) to outbound (in Compton).

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